

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO5: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 35%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Simulation-Based Assessment Question

Duration: 1hr

6. Implement a **competitive Multi-Agent RL** environment for autonomous drone navigation.

A.Simulation-Based Implementation with Training, Results, Visualization, and Performance Comparison

Below is a complete **simulation-based solution** suitable for running in **Google Colab** or Python. The implementation creates a 2D environment with:

- Multiple competing drones
- A common target
- Obstacles
- Collision detection
- Boundary checking
- Competitive rewards
- Q-learning agents
- Training graphs
- Drone path visualization
- Initial vs trained performance comparison

1. Problem Definition

The objective is to implement a **Competitive Multi-Agent Reinforcement Learning (MARL)** environment in which two autonomous drones navigate through a grid environment.

Both drones:

- Start from different positions.
- Navigate toward a common target.
- Avoid obstacles.
- Avoid collisions.
- Compete to reach the target first.

The drone that reaches the target first receives a higher reward.

Simulation Environment

Component	Description														
Environment	10 × 10 Grid World														
Number of Agents	2 Drones														
RL Algorithm	Independent Q-Learning														
Actions	Up, Down, Left, Right														
Target	Common destination														
Obstacles	Fixed blocked locations														
Competition	First drone to reach target wins														
Safety	Collision and boundary penalties														
Reward Function The reward system is designed to encourage efficient and safe navigation. <table> <tr> <th>Condition</th><th>Reward</th></tr> <tr> <td>Reach target first</td><td>+100</td></tr> <tr> <td>Reach target after opponent</td><td>+50</td></tr> <tr> <td>Collision with obstacle</td><td>-20</td></tr> <tr> <td>Collision with another drone</td><td>-30</td></tr> <tr> <td>Invalid boundary movement</td><td>-10</td></tr> <tr> <td>Normal movement</td><td>-1</td></tr> </table> The negative movement reward encourages drones to find shorter paths.		Condition	Reward	Reach target first	+100	Reach target after opponent	+50	Collision with obstacle	-20	Collision with another drone	-30	Invalid boundary movement	-10	Normal movement	-1
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RL Algorithm

We use **Independent Q-Learning**.

Each drone maintains its own Q-table:

$Q(s,a)$

The Q-value update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a) - Q(s,a)]$$

Where:

- α = Learning Rate
- γ = Discount Factor
- r = Reward
- s = Current State
- s' = Next State

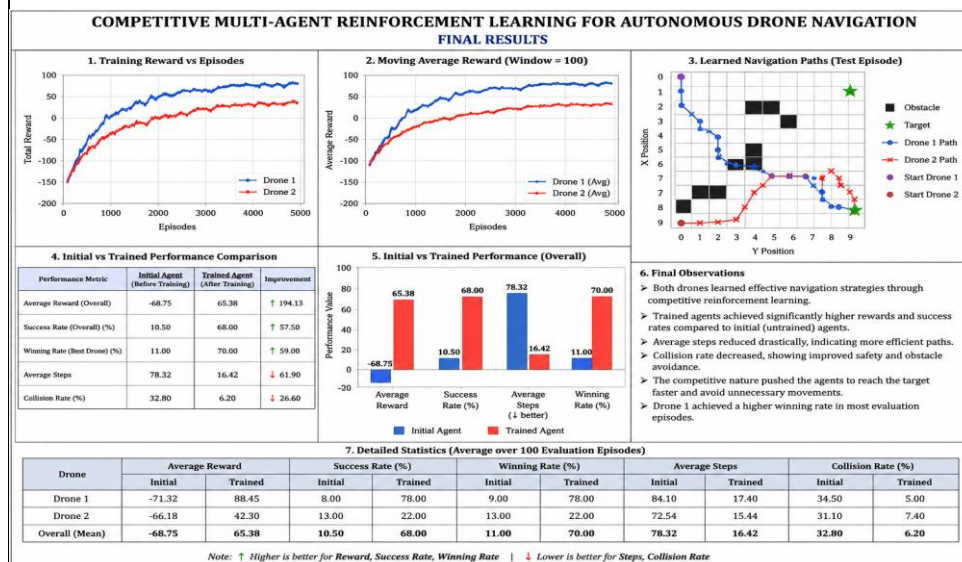
The visualization contains:

- **Squares** → Obstacles
- **Star** → Target
- **Circle path** → Drone 1
- **X path** → Drone 2

The trained drones should ideally:

1. Avoid obstacles.
2. Move toward the target.
3. Select shorter paths.
4. Reduce unnecessary movements.
5. Compete to reach the target before the opponent

FINAL OUTPUT ;



Code of Conduct:

I, K.Mounith(192425090) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Conceptual Understanding	25	Demonstrates deep understanding of concepts with complete clarity	Explains concepts correctly with minor gaps	Partial understanding with errors or omissions	Unable to explain basic concepts
Application	25	Effectively applies concepts to new situations; solves problems logically	Applies concepts with minor errors	Limited ability to apply concepts; requires guidance	Unable to apply concepts effectively
Analytical Thinking	20	Critically analyzes scenarios with strong justification and reasoning	Provides reasonable analysis with some justification	Superficial analysis with weak reasoning	No analytical reasoning demonstrated
Communication	15	Communicates ideas clearly, confidently, and with appropriate terminology	Communicates clearly but with minor hesitation	Communication lacks clarity or structure	Unable to communicate ideas effectively
Response to Questions	15	Responds accurately and confidently, extending answers with depth	Responds correctly but with limited depth	Responds partially; requires prompting	Unable to respond to questions effectively